

4.1.)

$$z = ax + by + c$$

$$\sum_i (z_i - (ax_i + by_i + c))^2$$

Radim
TrilskýPro lin. regresi v $\mathbb{R}^3$ dostaneme:

$$\sum z_i = a \sum x_i + b \sum y_i + c n$$

$$\sum x_i z_i = a \sum x_i^2 + b \sum x_i y_i + c \sum x_i$$

$$\sum y_i z_i = a \sum x_i y_i + b \sum y_i^2 + c \sum y_i$$

2 body: $(0,0,1), (1,0,1), (0,1,1), (1,1,2), (1,-1,0), (-1,1,1)$ dostaneme: $n = 6$

$$\sum x = 2, \sum y = 2, \sum z = 6$$

$$\sum x^2 = 4, \sum y^2 = 4$$

$$\sum xy = -1, \sum xz = 2, \sum yz = 4$$

Pozadíme do soustavy: $6 = 2a + 2b + 6c$

$$2 = 4a - b + 2c$$

$$4 = -a + 4b + 2c$$

K čehož dostaneme: $a + b + 3c = 3, 4a - b + 2c = 2, -a + 4b + 2c = 4$

$$\Rightarrow a = \frac{2}{5}, b = \frac{4}{5}, c = \frac{3}{5}$$

Hledaná rovina:

$$z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$